Bihar Engineering University, Patna

B.Tech 7th Semester Examination, 2024

Course : B.Tech
Code : 104701

Subject: Business Analytics

Time: 03 Hours
Full Marks: 70

Instructions:-

(i) The marks are indicated in the right-hand margin.
*(ii) There are **NINE** questions in this paper.*
*(iii) Attempt **FIVE** questions in all.*
*(iv) Question No. **1** is compulsory.*

Q.1 *Choose the correct answer of the following (any seven question only):* [2 x 7 = 14]

(a) What is the main focus of Descriptive Analytics?
(i) Predicting future trends
(ii) Understanding historical data patterns and summarizing insights
(iii) Prescribing actionable solutions
(iv) Building optimization models

(b) Simple Linear Regression is used to:
(i) Model nonlinear relationships
(ii) Determine the relationship between one dependent variable and one independent variable
(iii) Analyse categorical data
(iv) Identify clustering patterns

(c) Which of the following is an example of a dependent variable in regression analysis?
(i) Advertising budget
(ii) Sales revenue
(iii) Number of employees
(iv) Product cost

(d) What is a common tool used in Business Analytics technology for visualization?
(i) Microsoft Excel
(ii) Tableau
(iii) Apache Hadoop
(iv) SQL Server

(e) Which of the following is considered a key resource in Business Analytics?
(i) Office location
(ii) Employee attendance records
(iii) Company branding
(iv) Data technology and skilled personnel

(f) Which of the following is an example of a qualitative forecasting technique?
(i) Time series analysis
(ii) Delphi method
(iii) Regression analysis
(iv) Moving averages

(g) Which platform is often used for Monte Carlo Simulation in Business Analytics?
(i) Microsoft Excel
(ii) Analytic Solver Platform
(iii) Tableau
(iv) Hadoop

(h) Which step is NOT part of the Business Analytics process?
(i) Data collection
(ii) Data analysis
(iii) Data Modelling
(iv) Filing taxes

(i) Nonlinear optimization is often used in Prescriptive Analytics to:
(i) Simplify data visualizations
(ii) Analyse customer satisfaction
(iii) Solve complex decision-making problems
(iv) Store data more efficiently

(j) Data journalism focuses on:
(i) Visualizing only qualitative data
(ii) Automating storytelling processes
(iii) Using data to create engaging, fact-based stories for audiences
(iv) Analysing news trends only

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Q.2	(a)	Define forecasting and explain its significance in business decision-making	[7]
	(b)	What are decision trees, and how do they assist in evaluating multiple decision options? Provide an example.	[7]
Q.3	(a)	What is the Cash Budget Model? Discuss its role in managing financial risks.	[7]
	(b)	What are some common challenges in using regression analysis in Business Analytics?	[7]
Q.4	(a)	Explain how the Business Analytics process aligns with organizational structure.	[7]
	(b)	Explain the key elements of designing an effective information policy for Business Analytics.	[7]
Q.5	(a)	How does the Analytic Solver Platform enhance Monte Carlo Simulation in business decision-making?	[7]
	(b)	Explain the concept of data storytelling and its role in turning data insights into actionable business strategies.	[7]
Q.6	(a)	Define Business Analytics	[4]
	(b)	Discuss the importance of nonlinear optimization in Prescriptive Analytics with a real-world example.	[10]
Q.7		Discuss the scope of Business Analytics with examples of its applications in business decision-making.	[14]
Q.8		Explain forecasting time series with seasonality, and provide an example of its application in a business context	[14]
Q.9		Write short notes on ***any two*** of the following:-	[7 x 2 = 14]
	(a)	collaborative business intelligence	
	(b)	embedded business intelligence	
	(c)	Data Journalism	
	(d)	Overbooking Model	

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